5 Divided and Finite Differences - Continued

Properties of divided differences

Theorem 5.1. *Divided differences have a number of special properties that can simplify work with them:*

a)

$$f[x_0, x_1, \ldots, x_n] \;=\; \sum_{i=0}^{n} \frac{f(x_i)}{u'(x_i)} \;=\sum_{i=0}^{n} \frac{f(x_i)}{u_i(x_i)}, \tag{5.1}$$

where $u(x) = (x - x_0)(x - x_1)\ldots(x - x_n)$ *and* $u_i(x) = \dfrac{u(x)}{x - x_i}$.

b) *For any permutation* $\{i_0, i_1, \ldots, i_n\}$ *of the integers* $\{0, 1, \ldots, n\}$,

$$f[x_{i_0}, x_{i_1}, \ldots, x_{i_n}] \;=\; f[x_0, x_1, \ldots, x_n]. \tag{5.2}$$

c)

$$f[x_0, x_1, \ldots, x_n] \;=\; \frac{(Wf)(x_0, x_1, \ldots, x_n)}{V(x_0, x_1, \ldots, x_n)}, \tag{5.3}$$

where

$$Wf(x_0, x_1, \ldots, x_n) \;=\; \begin{vmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^{n-1} & f(x_0) \\ 1 & x_1 & x_1^2 & \ldots & x_1^{n-1} & f(x_1) \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^{n-1} & f(x_n) \end{vmatrix} \text{ and}$$

$$V(x_0, x_1, \ldots, x_n) = \begin{vmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & \ldots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^n \end{vmatrix} = \prod_{0 \le j < i \le n} (x_i - x_j) \text{ is the Vandermonde determinant.}$$

d) *Let* $e_k(x) = x^k, k \ge 0$. *Then*

$$e_k[x_0, x_1, \ldots, x_n] \;=\; \begin{cases} 0, & k < n \\ 1, & k = n \end{cases}.$$

For a polynomial of degree k, $P_k = a_0 + a_1 x + \cdots + a_k x^k$,

$$P_k[x_0, x_1, \ldots, x_n] \;=\; \begin{cases} 0, & k < n \\ a_n, & k = n \end{cases} . \tag{5.4}$$

e) *If $f \in C^n[a,b]$, where $[a,b]$ is the smallest interval containing the distinct nodes $\{x_0, \ldots, x_n\}$, then there exists $\xi_n \in (a,b)$ such that*

$$f[x_0, x_1, \ldots, x_n] \;=\; \frac{1}{n!} f^{(n)}(\xi_n), \tag{5.5}$$

(the mean-value formula for divided differences).

Remark 5.2. As a consequence of part e), if $f \in C^n[a,b]$ and $\alpha \in [a,b]$, then

$$\lim_{x_0,\ldots,x_n \to \alpha} f[x_0, x_1, \ldots, x_n] \;=\; \lim_{\xi_n \to \alpha} \frac{f^{(n)}(\xi_n)}{n!} \;=\; \frac{1}{n!} f^{(n)}(\alpha),$$

the computational formula for divided differences with multiple nodes.

5.2 Finite Differences

Definition 5.3. *Consider the equidistant nodes $x_i = x_0 + ih, i = 0, 1, \ldots, n,\ h > 0$ and denote by $f_i \;=\; f(x_i)$. The quantity*

$$\Delta^1 f(x_i) \;=\; f(x_{i+1}) - f(x_i) \;=\; f_{i+1} - f_i \tag{5.6}$$

*is called the **first-order forward difference** of f with step h at x_i, and*

$$\Delta^k f(x_i) \;=\; \Delta^{k-1} f(x_{i+1}) - \Delta^{k-1} f(x_i) \;=\; \Delta^{k-1} f_{i+1} - \Delta^{k-1} f_i \tag{5.7}$$

*is the k**th-order forward difference** of f with step h, at x_i.*

Remark 5.4.
1. As a convention, we use $\Delta^0 f(x_i) = f(x_i) = f_i$.
2. For easy computation (and implementation) of forward differences, we construct a *table of forward differences*, similar to the one used for divided differences, illustrated below for 4 nodes.

$$
\begin{array}{c|ccccccc}
x_0 & f_0 & \longrightarrow & \Delta f_0 & \longrightarrow & \Delta^2 f_0 & \longrightarrow & \Delta^3 f_0 \\
 & & \nearrow & & \nearrow & & \nearrow & \\
x_1 & f_1 & \longrightarrow & \Delta f_1 & \longrightarrow & \Delta^2 f_1 & & \\
 & & \nearrow & & \nearrow & & & \\
x_2 & f_2 & \longrightarrow & \Delta f_2 & & & & \\
 & & \nearrow & & & & & \\
x_3 & f_3 & & & & & &
\end{array}
$$

In a similar way, we define the **backward difference** ∇ by

$$
\begin{aligned}
\nabla^0 f_i &= f_i, \\
\nabla^1 f_i &= f_i - f_{i-1}, \\
\nabla^k f_i &= \nabla^{k-1} f_i - \nabla^{k-1} f_{i-1},
\end{aligned}
\tag{5.8}
$$

and they can also be easily computed in a table.

$$
\begin{array}{c|ccccccc}
x_0 & f_0 & & & & & & \\
 & & \searrow & & & & & \\
x_1 & f_1 & \longrightarrow & \nabla f_1 & & & & \\
 & & \searrow & & \searrow & & & \\
x_2 & f_2 & \longrightarrow & \nabla f_2 & \longrightarrow & \nabla^2 f_2 & & \\
 & & \searrow & & \searrow & & \searrow & \\
x_3 & f_3 & \longrightarrow & \nabla f_3 & \longrightarrow & \nabla^2 f_3 & \longrightarrow & \nabla^3 f_3
\end{array}
$$

Remark 5.5. These differences are referred to collectively as *finite differences*. Usually, if nothing is specified, by "finite" differences we mean "forward" differences.

Denote by

$$
X = \{x_i \mid x_i = x_0 + ih,\ i = \overline{0, n}, x_0, h \in \mathbb{R}\}
$$

and for $f : X \to \mathbb{R}$, by $f_i = f(x_i)$.

Let us write a few finite differences and notice a pattern.

$$\begin{aligned}
\Delta^1 f(x_i) &= f_{i+1} - f_i, \\
\Delta^2 f(x_i) &= \Delta^1 f_{i+1} - \Delta^1 f_i \;=\; f_{i+2} - f_{i+1} - \big(f_{i+1} - f_i\big) \;=\; f_{i+2} - 2f_{i+1} + f_i, \\
\Delta^3 f(x_i) &= \Delta^2 f_{i+1} - \Delta^2 f_i \;=\; f_{i+3} - 2f_{i+2} + f_{i+1} - \big(f_{i+2} - 2f_{i+1} + f_i\big) \\
&= f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i.
\end{aligned}$$

It can easily be proved (by induction) that

$$\Delta^n f(x_i) \;=\; \sum_{k=0}^{n} (-1)^k \binom{n}{k} f_{n-k+i},$$

or, equivalently, by the symmetry of combinations,

$$\Delta^n f(x_i) \;=\; \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f_{k+i}.$$

In particular, we have

$$\Delta^n f(x_0) \;=\; \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f_k. \tag{5.9}$$

Finite and divided differences for equally spaced nodes are closely related.

Proposition 5.6. *Let $f : X \to \mathbb{R}$. Then*

$$f[a, a+h, \ldots, a+nh] \;=\; \frac{1}{n!h^n} \Delta^n f(a). \tag{5.10}$$

Chapter 2. Numerical Solution of Systems of Linear Algebraic Equations

Systems of simultaneous linear equations occur in solving problems in a wide variety of disciplines, including Mathematics, Statistics, physical, biological and social sciences, engineering, business and many more. They arise directly in solving real-world problems, and they also occur as part of the solution process for other problems. Numerical solutions of boundary and initial value problems for differential equations are a rich source of linear systems, especially large-size ones.

In this chapter, we will examine the following problem: given a matrix $A \in \mathbb{R}^{n\times n}$ and a vector $b \in \mathbb{R}^n$, find $x \in \mathbb{R}^n$ such that

$$Ax \;=\; b.$$

There are two types of methods for the solution of algebraic linear systems:

- *direct (exact)* methods, that provide a solution in a finite number of steps (e.g., Cramer, Gaussian elimination, factorizations);
- *iterative* methods, which approximate the solution by a sequence converging to it (e.g., Jacobi, Gauss-Seidel, SOR).

1 Direct Methods

1.1 Gaussian Elimination

A linear system is easy to solve when the matrix of the system is *triangular*:

Definition 1.1. *A matrix $A = [a_{ij}]_{i,j=\overline{1,n}}$ is called*

- ***upper triangular**, if $a_{ij} \;=\; 0, \forall i \;>\; j$,*

$$A \;=\; \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ & a_{22} & \dots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & a_{nn} \end{bmatrix}, \tag{1.1}$$

- ***lower triangular***, *if* $a_{ij} = 0, \forall i < j$,

$$A = \begin{bmatrix} a_{11} & & & 0 \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, \tag{1.2}$$

- ***diagonal***, *if it is both upper and lower triangular,* $a_{ij} = 0, \forall i \neq j$,

$$A = \begin{bmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix}. \tag{1.3}$$

Remark 1.2. The determinant of an upper or lower triangular matrix is equal to the product of its diagonal elements

$$\det(A) = a_{11}a_{22}\dots a_{nn}.$$

So, an upper or lower triangular matrix is nonsingular if and only if *all* of its diagonal entries are nonzero.

Example 1.3. Solve the triangular systems

a)

$$\begin{bmatrix} 2 & 4 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} x = \begin{bmatrix} 8 \\ 0 \\ -1 \end{bmatrix}, \tag{1.4}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 1/2 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 8 \\ 4 \\ 3 \end{bmatrix}. \tag{1.5}$$

Solution.

a) The upper triangular system is

$$\left\{\begin{array}{rcrcrcr} 2x_1 & + & 4x_2 & + & 2x_3 & = & 8 \\ & & -x_2 & + & x_3 & = & 0 \\ & & & & -x_3 & = & -1 \end{array}\right.$$

We start from the bottom (the last equation) and solve recursively for each unknown:

$$\begin{aligned} x_3 &= \frac{-1}{-1} = 1, \\ x_2 &= \frac{1}{-1}\big(0 - x_3\big) = 1, \\ x_1 &= \frac{1}{2}\big(8 - 4x_2 - 2x_3\big) = 1. \end{aligned}$$

We found the solution

$$x = [1 \;\; 1 \;\; 1]^T.$$

b) For the lower triangular system:

$$\left\{\begin{array}{rcrcrcr} x_1 & & & & & = & 8 \\ 1/2x_1 & + & x_2 & & & = & 4 \\ 1/2x_1 & + & x_2 & + & x_3 & = & 3 \end{array}\right.$$

we start from the top and solve each equation going down:

$$\begin{aligned} x_1 &= \frac{8}{1} = 8, \\ x_2 &= \frac{1}{1}\Big(4 - \frac{1}{2}x_1\Big) = 0, \\ x_3 &= \frac{1}{1}\Big(3 - \frac{1}{2}x_1 - x_2\Big) = -1. \end{aligned}$$

So the solution is

$$x = [8 \;\; 0 \;\; -1]^T.$$

■

So, in general, for a nonsingular upper triangular matrix U, the system $Ux = b$ is easily solved by **backward substitution**:

$$\begin{aligned} x_n &= \frac{b_n}{u_{nn}}, \\ x_i &= \frac{1}{u_{ii}}\Big(b_i - \sum_{j=i+1}^{n} u_{ij}x_j\Big),\ i = \overline{n-1, 1} \end{aligned} \tag{1.6}$$

and if the nonsingular matrix L is lower triangular, then the system $Lx = b$ is solved by **forward substitution**:

$$\begin{aligned} x_1 &= \frac{b_1}{l_{11}}, \\ x_i &= \frac{1}{l_{ii}}\Big(b_i - \sum_{j=1}^{i-1} l_{ij}x_j\Big),\ i = \overline{2, n}. \end{aligned} \tag{1.7}$$

Gaussian elimination is a procedure for transforming a system into an equivalent (upper) triangular one, by doing the following elementary row operations:

- multiplying a row (equation) by a constant $\lambda \neq 0$,

$$(\lambda R_i) \ \rightarrow\ (R_i),$$

- multiplying a row by a constant $\lambda \neq 0$ and adding it to another row,

$$(R_i + \lambda R_j) \ \rightarrow\ (R_i),$$

- interchanging (permuting) two rows,

$$(R_i) \ \longleftrightarrow\ (R_j).$$

All these elementary operations are performed on the *augmented (extended)* matrix of the system

$$\widetilde{A} \ = \ [A \mid b] \ = \ \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & a_{1,n+1} \\ a_{21} & a_{22} & \dots & a_{2n} & a_{2,n+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & a_{n,n+1} \end{array}\right], \tag{1.8}$$

where $a_{i,n+1} = b_i,\ i = \overline{1, n}$.

Gaussian elimination goes as follows:

Assuming $a_{11} \neq 0$, at the first step, we eliminate (make 0) the coefficients of x_1 from every row below, i.e., every $R_j, j = \overline{2, n}$, using a_{11}, i.e. by

$$\left(R_j - \frac{a_{j1}}{a_{11}} R_1\right) \ \to \ (R_j).$$

Then we proceed the same for the coefficients of each $x_i, i = \overline{2, n-1}, j = \overline{i+1, n}$. This way we obtain a finite sequence of augmented matrices

$$\widetilde{A}^{(1)}, \ \widetilde{A}^{(2)}, \ \ldots, \ \widetilde{A}^{(n)},$$

where $\widetilde{A}^{(1)} \ = \ \widetilde{A}$ and (at step k) $\widetilde{A}^{(k)} \ = \ \left[a_{ij}^{(k)}\right]$ obtained by

$$\left(R_i - \frac{a_{i,k-1}^{(k-1)}}{a_{k-1,k-1}^{(k-1)}} R_{k-1}\right) \ \to \ (R_i).$$

Here, we denoted by $a_{ij}^{(l)}$ the (i, j) entry at step l. . The quantities

$$m_{i,k-1} \ = \ \frac{a_{i,k-1}^{(k-1)}}{a_{k-1,k-1}^{(k-1)}}, \ i = k, \ldots, n \tag{1.9}$$

are called *multipliers*. For equations $i = k, \ldots, n$, we subtract $m_{i,k-1}$ times E_{k-1} from E_i, eliminating x_{k-1} from E_i. The new coefficients and the right-hand sides in equations E_k through E_n are defined by

$$\begin{aligned} a_{ij}^{(k)} &= a_{ij}^{(k-1)} - m_{i,k-1} a_{kj}^{(k-1)}, \ i, j = k, \ldots, n, \\ b_i^{(k)} &= b_i^{(k-1)} - m_{i,k-1} b_k^{(k-1)}, \ i = k, \ldots, n. \end{aligned}$$

At every step k, the system corresponding to the augmented matrix $\widetilde{A}^{(k)}$ is equivalent to the original linear system (meaning, it has the same solution) and in it, the variable x_{k-1} was eliminated from the equations $E_k, E_{k+1}, \ldots, E_n$. Then the system corresponding to $\widetilde{A}^{(n)}$ is an equivalent upper triangular one:

$$\left\{ \begin{array}{ccccccccc} a_{11}^{(1)} x_1 & + & a_{12}^{(1)} x_2 & + & \ldots & + & a_{1n}^{(1)} x_n & = & a_{1,n+1}^{(1)} \\ & + & a_{22}^{(2)} x_2 & + & \ldots & + & a_{2n}^{(2)} x_n & = & a_{2,n+1}^{(2)} \\ & & & \ddots & & & \vdots & & \\ & & & & & & a_{nn}^{(n)} x_n & = & a_{n,n+1}^{(n)} \end{array} \right. , \tag{1.10}$$

which is solved by backward substitution (1.6).

Of course, in all this we need $a_{ii}^{(i)} \ \neq \ 0$.The element $a_{ii}^{(i)}$ is called **pivot**. If at any time during the elimination process, we find $a_{kk}^{(k)} \ = \ 0$, then we look further down in that column for a pivot, i.e., we interchange rows

$$(R_k) \ \longleftrightarrow \ (R_p),$$

where p is the smallest integer $k+1 \leq p \leq n$ with $a_{pk}^{(k)} \neq 0$. If the original system is nonsingular, it can be shown that one of the equations following E_k *must* contain a term involving x_k with a nonzero coefficient, so it is always possible to find a pivot.

In fact, in practice, when implementing Gaussian elimination, pivoting is necessary even if the pivot is *not* zero, but small, compared to the rest of the elements in that column. We should avoid using coefficients that are nearly zero as pivot elements, because such a pivot can produce substantial rounding errors and even cancellations. Instead, we can use several types of *pivoting*.

- We can choose the pivot to be the largest element (in absolute value) in that column, below the main diagonal, i.e.

$$\left|a_{pk}^{(k)}\right| \;=\; \max_{k \leq l \leq n} \left|a_{lk}^{(k)}\right|. \tag{1.11}$$

  In most instances, it decreases the propagated effects of rounding errors. With partial pivoting, the multipliers $m_{i,k}$ in (1.9) will satisfy

$$|m_{i,k}| \;\leq\; 1,\; 1 \leq k < i \leq n.$$

  This will help reduce loss-of-significance errors, because multiplications by m_{ik} will not lead to much larger numbers.
  This is called **partial pivoting (maximal pivoting on columns)** and it is the most popular one in practice.

- We can do **scaled pivoting on columns**: First, we define a *scaling factor* for each row

$$s_i \;=\; \max_{j=\overline{1,n}} |a_{ij}| \;\text{ or } s_i \;=\; \sum_{j=1}^{n} |a_{ij}|,\; i = \overline{1,n}.$$

  If there exists an i such that $s_i = 0$, then the matrix is singular. For a nonsingular matrix, we use the scaling factor to choose the pivot. At each step i, we find the smallest $p,\; i \leq p \leq n$ such that

$$\frac{|a_{pi}|}{s_i} \;=\; \max_{1 \leq j \leq n} \frac{|a_{ji}|}{s_j} \tag{1.12}$$

  and then interchange rows $(R_i) \;\longleftrightarrow\; (R_p)$ so that the pivot is a_{pi}. This ensures the fact that the maximal element in each column has the relative size 1, before we compare and interchange rows. Also, dividing by the scaling factor does not produce any extra rounding

errors.

- The third method is **total (maximal) pivoting**. At each step k, we find

$$|a_{pq}| \;=\; \max\{|a_{ij}|, i,j = \overline{k,n}\} \tag{1.13}$$

and interchange both the rows and the columns,

$$(R_k) \longleftrightarrow (R_p), \;\; (C_k) \longleftrightarrow (C_q).$$

But then we have to keep track of the columns (unknowns) interchanges.

Remark 1.4. If A is singular of rank $p-1$, then at step p we get

$$\widetilde{A}^{(p)} \;=\; \left[\begin{array}{ccccccc|c} a_{11}^{(1)} & a_{12}^{(1)} & \dots & a_{1,p-1}^{(1)} & a_{1p}^{(1)} & \dots & a_{1n}^{(1)} & a_{1,n+1}^{(1)} \\ 0 & a_{22}^{(2)} & \dots & a_{2,p-1}^{(2)} & a_{2p}^{(2)} & \dots & a_{2n}^{(2)} & a_{2,n+1}^{(2)} \\ \vdots & & \ddots & \vdots & \vdots & & \vdots & \vdots \\ \vdots & & & a_{p-1,p-1}^{(p-1)} & a_{p-1,p}^{(p-1)} & & a_{p-1,n}^{(p-1)} & a_{p-1,n+1}^{(p-1)} \\ \vdots & & & & 0 & \dots & 0 & a_{p,n+1}^{(p)} \\ \vdots & & & & & \ddots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & \dots & \dots & 0 & a_{n,n+1}^{(n)} \end{array}\right].$$

So, if $a_{i,n+1}^{(i)} \;=\; b_i^{(i)} \;=\; 0$, for *all* $i = p, p+1, \dots, n$, then the system is compatible, but undetermined (i.e., it has an infinite number of solutions), otherwise, the system is incompatible (no solution). Thus, Gaussian elimination can also be used to discuss the solvability of the linear system.

Example 1.5. Solve the system

$$\left\{\begin{array}{rcrcrcr} x_1 & - & x_2 & + & x_3 & = & -1 \\ -2x_1 & + & 2x_2 & + & x_3 & = & 2 \\ -3x_1 & - & x_2 & + & 5x_3 & = & -5 \end{array}\right. ,$$

by Gaussian elimination with different types of pivoting.

Solution. The augmented matrix of the system is

$$\widetilde{A} \;=\; \left[\begin{array}{rrr|r} 1 & -1 & 1 & -1 \\ -2 & 2 & 1 & 2 \\ -3 & -1 & 5 & -5 \end{array}\right],$$

Partial pivoting

On the first column, the largest element in absolute value is -3, so that will be the pivot. Thus, first we interchange $(R_1) \longleftrightarrow (R_3)$. We get

$$\widetilde{A} \sim \left[\begin{array}{rrr|r} \boxed{-3} & -1 & 5 & -5 \\ -2 & 2 & 1 & 2 \\ 1 & -1 & 1 & -1 \end{array}\right] \underset{\sim}{\begin{array}{r} (-\frac{2}{3}\,R_1 + R_2) \;\to\; (R_2) \\ (\frac{1}{3}\,R_1 + R_3) \;\to\; (R_3) \end{array}} \left[\begin{array}{rrr|r} -3 & -1 & 5 & -5 \\ 0 & 8/3 & -7/3 & 16/3 \\ 0 & -4/3 & 8/3 & -8/3 \end{array}\right]$$

Further, we have

$$\widetilde{A} \sim \left[\begin{array}{rrr|r} -3 & -1 & 5 & -5 \\ 0 & \boxed{8/3} & -7/3 & 16/3 \\ 0 & -4/3 & 8/3 & -8/3 \end{array}\right] \underset{\sim}{(\tfrac{1}{2}\,R_2 + R_3) \;\to\; (R_3)} \left[\begin{array}{rrr|r} -3 & -1 & 5 & -5 \\ 0 & 8/3 & -7/3 & 16/3 \\ 0 & 0 & 3/2 & 0 \end{array}\right]$$

Now we solve by back substitution (1.6), to get

$$x \;=\; [1\;\;2\;\;0]^T.$$

Scaled partial pivoting

We compute the scaling factors using sums on each row. At the first step $k = 1$, we get

$$\begin{aligned} s &= [3,\ 5,\ 9] \\ \left[\frac{|a_{j,1}|}{s_j}\right] &= [1/3,\ 2/5,\ 3/9] \;=\; [5/15,\ 6/15,\ 5/15],\ j = 1, 2, 3. \end{aligned}$$

The maximum of the three fractions is the second, so $p = 2$. We interchange $(R_1) \longleftrightarrow (R_2)$ and make zeros below it.

$$\widetilde{A} \sim \left[\begin{array}{rrr|r} \boxed{-2} & 2 & 1 & 2 \\ 1 & -1 & 1 & -1 \\ -3 & -1 & 5 & -5 \end{array}\right] \underset{\sim}{\begin{array}{r} (\frac{1}{2}\,R_1 + R_2) \;\to\; (R_2) \\ (-\frac{3}{2}\,R_1 + R_3) \;\to\; (R_3) \end{array}} \left[\begin{array}{rrr|r} -2 & 2 & 1 & 2 \\ 0 & 0 & 3/2 & 0 \\ 0 & -4 & 7/2 & -8 \end{array}\right]$$

At step $k = 2$, obviously, $p = 3$, so we interchange $(R_2) \longleftrightarrow (R_3)$ to get

$$\widetilde{A} \sim \left[\begin{array}{rrr|r} -2 & 2 & 1 & 2 \\ 0 & -4 & 7/2 & -8 \\ 0 & 0 & 3/2 & 0 \end{array}\right]$$

and we are done. By back substitution we get the (obviously, same) solution

$$x \;=\; [1\;\;2\;\;0]^T.$$

Total pivoting

At step $k = 1$, since

$$\max_{i,j=\overline{1,3}} |a_{ij}| \;=\; 5 \;=\; |a_{33}|,$$

we interchange *both* rows and columns, $(R_1) \longleftrightarrow (R_3),\ (C_1) \longleftrightarrow (C_3)$, to get

$$\widetilde{A} \;=\; \left[\begin{array}{rrr|r} 1 & -1 & 1 & -1 \\ -2 & 2 & 1 & 2 \\ -3 & -1 & 5 & -5 \end{array}\right] \sim \left[\begin{array}{rrr|r} -3 & -1 & 5 & -5 \\ -2 & 2 & 1 & 2 \\ 1 & -1 & 1 & -1 \end{array}\right] \sim \left[\begin{array}{rrr|r} 5 & -1 & -3 & -5 \\ 1 & 2 & -2 & 2 \\ 1 & -1 & 1 & -1 \end{array}\right],$$

which is now a system for the *new unknown* $x' \;=\; [x_3\;x_2\;x_1]^T$. We proceed to make zeros on the first column below the diagonal.

$$\widetilde{A} \;\sim\; \left[\begin{array}{rrr|r} \boxed{5} & -1 & -3 & -5 \\ 1 & 2 & -2 & 2 \\ 1 & -1 & 1 & -1 \end{array}\right] \underset{\sim}{\begin{array}{c} (-\frac{1}{5}\,R_1 + R_2) \;\to\; (R_2) \\ (-\frac{1}{5}\,R_1 + R_3) \;\to\; (R_3) \end{array}} \left[\begin{array}{rrr|r} 5 & -1 & -3 & -5 \\ 0 & 11/5 & -7/5 & 3 \\ 0 & -4/5 & 8/5 & 0 \end{array}\right]$$

At step $k = 2$,

$$\max_{i,j=2,3} |a_{ij}| \;=\; \frac{11}{5} \;=\; |a_{22}|,$$

so no (row or column) interchanges are necessary. We have

$$\widetilde{A} \;\sim\; \left[\begin{array}{rrr|r} 5 & -1 & -3 & -5 \\ 0 & \boxed{11/5} & -7/5 & 3 \\ 0 & -4/5 & 8/5 & 0 \end{array}\right] \underset{\sim}{(\tfrac{4}{11}\,R_2 + R_3) \;\to\; (R_3)} \left[\begin{array}{rrr|r} 5 & -1 & -3 & -5 \\ 0 & 11/5 & -7/5 & 3 \\ 0 & 0 & 12/11 & 12/11 \end{array}\right]$$

By back substitution, we get

$$x' \;=\; [0\;\;2\;\;1]^T \text{ and (again) } x \;=\; [1\;\;2\;\;0]^T.$$

■

Remark 1.6.

1. The elements under the main diagonal (which become 0) need not be computed.

2. When pivoting, we do not need to *physically* interchange rows or columns. Just keep one (or two) permutation vector(s) p (q) with $p[i]$ $(q[j])$ meaning that the row (column) p (q) has been interchanged with row (column) i (j). This is especially a good solution if matrices are stored row by row or column by column.

3. Gaussian elimination can be used to find the inverse A^{-1} of a nonsingular matrix. For each $k = \overline{1,n}$, column k of A^{-1} can be found by solving the system $Ax = e_k$, where $\{e_k\}$ is the canonical basis of $\mathbb{R}^n$, $e_k = [0\ 0\ \ldots\ 0\ 1\ 0\ \ldots\ 0]^T$, with 1 on the kth slot. Alternatively, the inverse of A can be found by Gaussian elimination on the matrix

$$[A \mid I] \sim \ldots \sim [I \mid A^{-1}].$$

Computational Complexity

Since the time required for a certain algorithm to run depends on the details of the hardware used, it is more representative to count the number of some elementary operations, such as multiplications, divisions, additions, and subtractions. Strictly speaking, we should count separately additions/subtractions and multiplications/divisions, since the latter take slightly more time to be performed, but we will count everything together. Let us assess the computational cost of Gaussian elimination and compare it to other methods.

Gaussian elimination

At step $k = 1$, we perform $n - 1$ divisions, $(n-1)n$ multiplications and $(n-1)n$ additions, so a total of $2n(n-1) + (n-1)$ flops. At step $k = 2$, there are $2(n-1)(n-2) + (n-2)$ flops and so on until step $k = n - 1$. So, the actual elimination process requires

$$\sum_{k=1}^{n-1} \Big[2(n-k)(n-k+1) + (n-k)\Big] = \sum_{i=1}^{n-1} \Big[2i(i+1) + i\Big] = \frac{n(n-1)(4n+7)}{6}$$

flops. Back substitution adds another

$$1 + 3 + \cdots + 2n - 1 = \sum_{i=1}^{2n-1} i - 2\sum_{i=1}^{n-1} i = n^2$$

flops, for a total of

$$\frac{n(4n^2 + 9n - 7)}{6} = \mathcal{O}\Big(\frac{2}{3}n^3\Big)$$

operations.

Cramer's rule

Assume the determinants in Cramer's rule are computed using expansion by minors. That means that to solve an $n \times n$ system, we have to calculate $n+1$ determinants. If D_n denotes the number of elementary operations needed to compute the determinant of an $n \times n$ matrix, then

$$\begin{aligned}
D_2 &= 2+1=3 \text{ (two multiplications and one subtraction)}, \\
D_3 &= 3D_2+3+2 \text{ } (3D_2,\ 3 \text{ multiplications and 2 additions/subtractions}), \\
&\dots \\
D_n &= nD_{n-1}+n+n-1.
\end{aligned}$$

So,

$$D_n \ > \ nD_{n-1} \ > \ n(n-1)D_{n-2} \ > \ \dots \ > \ n(n-1)\dots 2D_1 \ = \ n!.$$

Then the operation count for Cramer's rule is

$$\mathcal{O}\big((n+1)!\big).$$

For example, for $n=10$, Gaussian elimination uses about 805 operations, while Cramer's rule uses around $3,628,800$ operations. This should emphasize the point that Cramer's rule is *not* a practical computational method, and that it should be considered as just a theoretical mathematics tool.

1.2 Factorization Based Methods

These are methods that use the fact that the matrix of coefficients of a linear system being solved can be *factored (decomposed)* into the product of two triangular matrices.

1.2.1 LU Factorization

Theorem 1.7. *If no row interchanges are necessary in the Gaussian elimination process for solving the system $Ax=b$, then A can be factored as*

$$A \ = \ LU, \tag{1.14}$$

where L and U are lower and upper triangular matrices, respectively. The pair (L,U) is called an LU ***factorization (decomposition)*** *of the matrix A.*

Sketch of Proof. The first step is to partition A as

$$A = \left[\begin{array}{c|ccc} a_{11} & a_{12} & \cdots & a_{1n} \\ \hline a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array}\right] = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix},$$

where v is a column vector of length $n-1$, w^* is a row vector of length $n-1$ and A' is an $(n-1)\times(n-1)$ matrix. Then we can factor A as

$$A = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ v/a_{11} & I_{n-1} \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & A' - vw^*/a_{11} \end{bmatrix}.$$

The matrix $A' - vw^*/a_{11}$ is called the **Schur complement** of A with respect to a_{11}.
Then, we proceed recursively:

$$A' - vw^*/a_{11} = L'U'.$$

So

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 \\ v/a_{11} & I_{n-1} \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & A' - vw^*/a_{11} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ v/a_{11} & I_{n-1} \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & L'U' \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ v/a_{11} & L' \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & U' \end{bmatrix} \end{aligned}$$

until we get a scalar (a 1×1 matrix) that can no longer be partitioned.

□

Remark 1.8. If $A = LU$, then solving the system $Ax = b$ is reduced to solving two triangular systems

$$\begin{aligned} Ly &= b \text{ and} \\ Ux &= y. \end{aligned} \tag{1.15}$$

Example 1.9. Use LU decomposition to solve the system

$$\begin{cases} 2x_1 + 4x_2 + 2x_3 = 8 \\ x_1 + x_2 + 2x_3 = 4 \\ x_1 + x_2 + x_3 = 3 \end{cases}$$

Solution. We have

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} = \left[\begin{array}{c|cc} 2 & 4 & 2 \\ \hline 1 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right],$$

so, at the first step,

$$a_{11} = 2, \;\; v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \;\; w^* = [4 \;\; 2], \;\; A' = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \;\; \left[\begin{array}{c|cc} 2 & 4 & 2 \\ \hline 1/2 & & \\ 1/2 & & \end{array}\right].$$

The first Schur complement is

$$A' - vw^*/a_{11} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}[4 \;\; 2] = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

and, for now, we have

$$\left[\begin{array}{c|cc} 2 & 4 & 2 \\ \hline 1/2 & -1 & 1 \\ 1/2 & -1 & 0 \end{array}\right] = \left[\begin{array}{c|c|c} 2 & 4 & 2 \\ \hline 1/2 & -1 & 1 \\ \hline 1/2 & -1 & 0 \end{array}\right] = \left[\begin{array}{c|c|c} 2 & 4 & 2 \\ \hline 1/2 & -1 & 1 \\ \hline 1/2 & 1 & \end{array}\right],$$

where 1 was obtained by dividing $\frac{-1}{-1}$.
The last Schur complement is

$$0 - (-1)/(-1) \cdot 1 = -1$$

and the final decomposition is

$$\left[\begin{array}{c|c|c} 2 & 4 & 2 \\ \hline 1/2 & -1 & 1 \\ \hline 1/2 & 1 & -1 \end{array}\right].$$

We take the upper triangular part (***including*** the main diagonal) for U and the lower triangular part (***without*** the main diagonal) for L (which will have all **1**'s on the main diagonal), to get

$$L = \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 1/2 & \mathbf{1} & 0 \\ 1/2 & 1 & \mathbf{1} \end{bmatrix}, \;\; U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

and check that indeed $A = LU$.

Now, solve $Ly = b = [8\ \ 4\ \ 3]^T$, which, from Example 1.3b) has solution $y = [8\ \ 0\ \ -1]^T$ and then $Ux = [8\ \ 0\ \ -1]^T$, which, from Example 1.3a), gives the solution

$$x = [1\ \ 1\ \ 1]^T.$$

Check that $Ax = b$. ■

Remark 1.10.

1. If all that is required is that L be lower and U be upper triangular, then the LU decomposition is *not* unique. We can make it unique by imposing more conditions. For instance, if we require $l_{ii} = 1, i = \overline{1,n}$, we have *Doolittle* factorization and if we impose $u_{ii} = 1, i = \overline{1,n}$, we get the *Crout* factorization. The procedure described in the proof of Theorem 1.7 leads to Doolittle factorization.

2. The matrix $U = [u_{ij}]$ in the Doolittle factorization is the upper triangular matrix obtained by Gaussian elimination (without pivoting),

$$u_{ij} = a_{i,j}^{(i)}, \ \ i \leq j, \tag{1.16}$$

while $L = [l_{ij}]$ is the matrix of the *multipliers*

$$l_{ij} = m_{ij} = \frac{a_{i,j}^{(j)}}{a_{j,j}^{(j)}}, \ i \geq j. \tag{1.17}$$

3. Examples of cases when no row interchanges are necessary:

- A is **diagonally dominant on rows**,

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|, \ i = \overline{1,n}. \tag{1.18}$$

- A is **positive definite**,

$$x^T A x > 0, \ \forall x \neq 0. \tag{1.19}$$